

NUS HPC-AI Hopper Cluster



NUS IT Research Computing

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Note



The system is new, unforeseen issues may arise



Please report any such issues on dedicated Teams channel

We have a
Microsoft
Teams
Team



HPC Priority Projects
User Channel

...

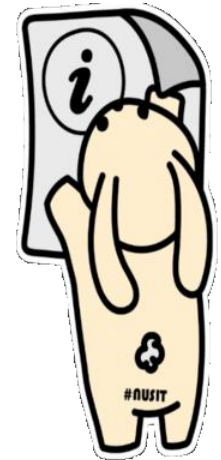
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Access

- Ensure you are on NUS Campus network or NUS VPN
- Option 1: Terminal Interface
 - ssh nus_id@hopper.nus.edu.sg

If you are unable to login on campus network, please connect to NUS VPN and try again.



Resources: Hardware

- 6 nodes * 8 H100-80GB GPUs, 2x 56 core Intel 8480+ CPUs ea., 2TB RAM ea.
- 1 dedicated login node [hopper.nus.edu.sg] ([Internet Access Available](#))
- GPUs are Hopper architecture
 - Take advantage of acceleration libraries like bitsandbytes, HuggingFace Accelerate, OpenAI Triton, unsloth, Megatron-LM, axolotl
- 8 GPUs per node
 - Take advantage of distributed training techniques/libraries:
 - Deepspeed, FSDP, HuggingFace Accelerate, axolotl
- [We have an existing GPU cluster -> Volta GPU Cluster](#)

Resources: Storage

Directories	Feature	Disk Quota	Backup	Description
/home/svu/\$USERID	Available across Hopper Cluster	40 GB	Snapshot	Home Directory. U:drive on your PC.
/hpctmp/\$USERID	Local on All Atlas/Volta/Hopper clusters	500 GB	No	Working Directory. Files older than 60 days are purged automatically
/local_ssd/\$USERID	Available individual Hopper nodes	Disk size 6.9TB	No	For quick read/write access to datasets. Create a folder with your NUSNET ID. Purged weekly.
/scratch/\$USERID	Available across Hopper Cluster	1TB	No	For quick read/write access to datasets. Create a folder with your NUSNET ID. Files older than 60 days are purged automatically

Resources: Containers

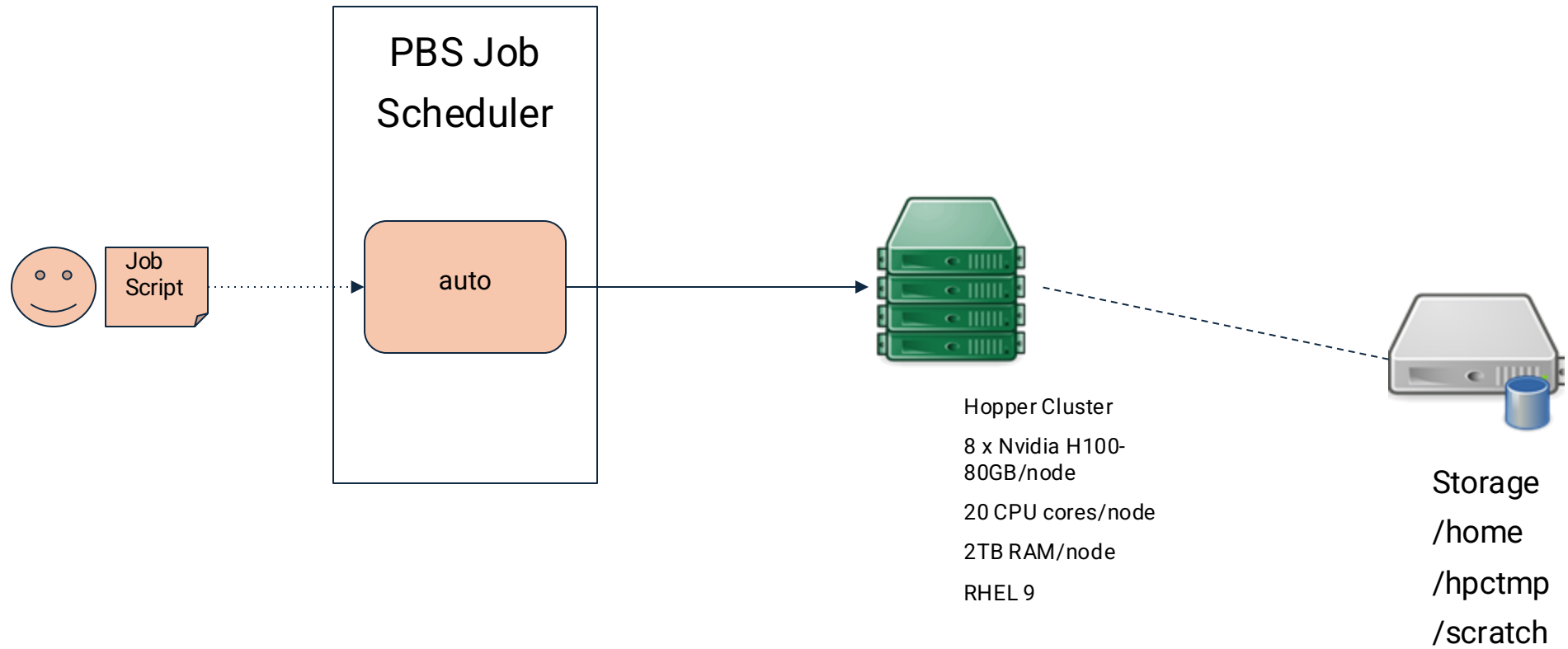
- All jobs must run using **Singularity/Apptainer Containers**
- We provide Singularity/Apptainer Containers that contain either
 - popular Deep Learning frameworks (e.g.: Pytorch, Tensorflow)
 - CUDA libraries
- If the containers we provide do not suit your needs, you can build your own Singularity/Apptainer containers or put in a request to get us to build it for you
- When building containers yourself, do ensure that **CUDA libraries** are installed and that the Deep Learning framework you install is the **GPU version**
- **Pre-built containers available at**
 - **/app1/common/singularity-img/hopper/**

Resources: Containers

- CUDA containers
 - Located in: **/app1/common/singularity-img/hopper/cuda**
 - Comes pre-installed with python and virtualenv
 - You can create virtualenvs (stored in /home or /scratch or /hpctmp) and install your desired packages (e.g.: Tensorflow/Pytorch/JAX/etc) instead of using the pre-built Pytorch or Tensorflow containers.
 - Alternatively, you can install packages using pip install --prefix with your desired folder for storing python packages
 - More details about python packaging in later slides

File Transfer

- MobaXterm built-in sftp client
- Filezilla client
- Linux/Mac OS/Windows Terminal Tools
 - scp
 - rsync
 - sftp



Projects

- Project Budgets have been allocated and enabled.
- Check your allocated project ID/name and consumption using
 - `'hpc project'`
- Set the Project name in the `#PBS -P Project-Name` field in your job script
 - `#PBS -P H1XXXXX`

```
[ccekwk@hopper-l-01 axolotl]$ hpc project

=====
Hopper H100 GPU-Hours/Credits for: ccekwk
=====
Project-Name | Total Credits | Net Balance
-----
[REDACTED] | 10000.0 | 9957.832
Users : sauyong,cceyeoeh,mcosta,rikky,ccewjh,ccekwk,ccev3422
-----
H100[REDACTED] | 1000.0 | 1000.0
Users : ccev3422,ccekwk,ccewjh
-----

#####
## Please quote the right Project-Name in below line in PBS job script, ##
## in order to use the GPU-Hours allocated in the Project. ##
#####
#PBS -P Project-Name
```

Jobs

- All jobs must be submitted as **batch jobs** to the job scheduler



Short Interactive jobs

Queue limitations:

Max wall time 2 hours

Max GPU 8 **[Ensure your script requires >1 GPU when requesting for >1 GPU]**

Min GPU 1

GPU in increments of 1

Interactive: Command Line Interface

1. Launch interactive job from Login node
 - `qsub -I -l select=1:ngpus=1 -l walltime=02:00:00`
 - You should see your prompt change to `nusnet_id@hopper-0X`
2. Run '`module load singularity`' to load singularity container runtime
3. Load your desired container:
 - `singularity exec -e /app1/common/singularity-img/hopper/pytorch/pytorch_2.3.0.sif bash`
4. Finally in the container you can run your scripts



Long Batch Jobs

Queue Limitations:

Max wall time:

- 6 days for 1 GPU jobs,
- 3 days for 8 GPU jobs,
- 2 days for multi-node jobs

Max GPU 8 [per Single-node] **[Ensure your script requires >1 GPU when requesting for >1 GPU]**

Min GPU 1

GPU in increments of 1

Sample Batch Job Script

```
#!/bin/bash
#PBS -P H1XXXXX # specify allocated project code
#PBS -j oe
#PBS -k oed
#PBS -N pytorch
#PBS -q auto
#PBS -l select=1:ngpus=8
#PBS -l walltime=24:00:00

cd $PBS_O_WORKDIR;

image="/app1/common/singularity-img/hopper/cuda/cuda_12.1.0-cudnn8-devel-u20.04.sif"
module load singularity

singularity exec -e $image bash << EOF > stdout.$PBS_JOBID 2> stderr.$PBS_JOBID
source /scratch/ccekwk/virtualenvs/axolotl/bin/activate

python cifar10_resnet.py

# you can put more commands here
# echo "Hello World"

EOF
```

Sample Batch Job Script (Custom Python Package Prefix)

```
#!/bin/bash
#PBS -P H1XXXXX # specify allocated project code
#PBS -j oe
#PBS -k oed
#PBS -N pytorch
#PBS -q auto
#PBS -l select=1:ngpus=1
#PBS -l walltime=24:00:00

cd $PBS_O_WORKDIR;

image="/app1/common/singularity-img/hopper/cuda/cuda_12.1.0-cudnn8-devel-u20.04.sif"
module load singularity

singularity exec -e $image bash << EOF > stdout.$PBS_JOBID 2> stderr.$PBS_JOBID
export PYTHONPATH=$PYTHONPATH:/home/svu/nusnetid/hopper_pypkg/cuda_12.1.0-cudnn8-devel-u20.04/local/lib/python3.10/dist-packages

python cifar10_resnet.py

# you can put more commands here
# echo "Hello World"

EOF
```



Job Scheduler

Job Submission

You have to run:

1. Prepare your **python script** in your working directory
2. **Create a PBS job script** and save it in your working directory
 - a. Example job scripts are in the following 2 slides
3. **Submit PBS job script** to PBS Job Scheduler

Server will run:

1. Job is in PBS Job Scheduler queue
2. Job Scheduler waits for server resources to be available
3. If available, Job Scheduler runs your script on remote gpu server

Job Status

Save your job script (previous slides for examples) in a text file (e.g. train.pbs) then run the following commands

```
shell$ qsub train.txt
675674.venus01
```

```
shell$ qstat -xfn
```

```
venus01:
```

Job ID	Username	Queue	Jobname	SessID	NDS	TSK	Req'd Memory	Req'd Time	S	Elap Time
669468.venus01	ccekwk	volta	cifar_noco	--	1	1	20gb	24:00	F	--
674404.venus01	ccekwk	volta	cifar_noco	--	1	1	20gb	24:00	F	--
675674.venus01	ccekwk	volta	cifar_noco	--	1	1	20gb	24:00	Q	--

Statuses: Q(ueue), F(inish), R(unning), E(nding), H(old)

Useful PBS Job Commands

Action	Command
Job submission	<code>qsub my_job_script.txt</code>
Job deletion	<code>qdel my_job_id</code>
Job listing (Simple)	<code>qstat</code>
Job listing (Detailed)	<code>qstat -ans1</code>
Queue listing	<code>qstat -q</code>
Completed Job listing	<code>qstat -H</code>
Completed and Current Job listing	<code>qstat -x</code>
Full info of a job	<code>qstat -f job_id</code>

gstat command - Queue Load Information

- Use the gstat command to find out about the queue load

```
[ccekwk@hopper-l-01 axolotl]$ gstat

-----
Last Update TIME: 20240815_15.20
-----

===== Status of Hopper GPU Queues =====
Queue :      Running :      Waiting : Users_in_Queue
auto: 0 : 13 : 4
-----

===== Status of Hopper GPU Cluster Nodes =====
TIME: Thu Aug 15 03:22:59 PM +08 2024
-----
```

node	state	njobs	run	susp	mem f/t	ncpus f/t	nmics f/t	ngpus f/t	jobs
hopper-06	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--
hopper-03	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--
hopper-02	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--
hopper-04	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--
hopper-01	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--
hopper-05	free	0	0	0	2tb/2tb	112/112	0/0	8/8	--

Python packages

Managing Python Packages

2 Options

CUDA
Containers

Using **CUDA containers** and
virtualenv/custom PYTHONPATH

Deep
Learning
Containers

Using Pytorch/Tensorflow/non-general
containers with pip install --prefix (custom
PYTHONPATH)

[Option 1] Using Virtualenv

- On **Hopper Login Node** (NO NEED TO LAUNCH JOB):

```
$ module load singularity
```

```
$ singularity exec -e /app1/common/singularity-img/hopper/cuda/cuda_12.1.0-cudnn8-devel-u20.04.sif bash
```

- Executing the above commands will bring you into the container.

In Container:

Setup virtualenv:

- `mkdir /hpctmp/[nusnetid]/virtualenvs`
- `python3.10 -m venv /hpctmp/[nusnetid]/virtualenvs/[envname]`

Install Packages in virtualenv:

- `source /hpctmp/[nusnetid]/virtualenvs/[envname]/bin/activate`
- `pip install package_name`

[Option 2] Installing Python Packages with Custom Path

On **Hopper Login Node** (NO NEED TO LAUNCH JOB):

```
$ module load singularity
```

```
$ singularity exec -e /app1/common/singularity-img/3.0.0/[yourcontainer.simg] bash
```

Executing the above commands will bring you into the container.

In Container:

```
pip install --prefix=/home/svu/[nusnet_id]/hopper_pypkg/[container_name]/  
package_name
```

For example name it as the container you are using.

*Replace [nusnet_id] with YOUR NUSNET ID.

*Replace [container_name] with the name of your container, excluding ".sif".

E.g.: /home/svu/nusnetid/hopper_pypkg/cuda_12.1.0-cudnn8-devel-u20.04

Follow the above naming format so that you can use Jupyter on HPC OnDemand.

[Option 2] Installing Python Packages: DON'Ts

- Don't install using `pip install --user`

[Option 2] Using User-installed Python Packages: Custom Path

In job script add the following:

If you used Pytorch container:

```
export  
PYTHONPATH=$PYTHONPATH:/home/svu/[nusnet_id]/hopper_pypkg/[container_name]/local/lib/python[3.10/3.11]/dist-  
packages
```

If you named volta_pypkg something else, please remember to change it

Replace [nusnet_id] with YOUR NUSNET ID

Replace [container_name] with the container name you are using, without “*.sif” extension.

Application Notes

- Set your **HuggingFace** cache to a path on /scratch
 1. `mkdir -p /scratch/nusnetid/cache`
 2. `export HF_HOME=/scratch/nusnetid/cache`
- Set **OpenAI Triton** cache to a path on /scratch
 1. `mkdir -p /scratch/nusnetid/cache`
 2. `export XDG_CACHE_HOME=/scratch/nusnetid/cache`
- Add the above export lines to your job script as well

Expectations

Green Computing Initiative

- Jobs must be able to fully utilise requested resources
 - E.g.: request 8 gpu for 1 job, application executed by job must be able to utilise **all 8 GPUs concurrently**.
- Resource utilisation will be monitored
- (Persistent) under utilization is highly discouraged

Acknowledgements for GPU Priority Program

- In your publications you must add the following acknowledgements:
 - Grant number: **NUSREC-HPC-00001**

Example:

“We would like to acknowledge that computational work involved in this research work is partially/fully supported by NUS IT’s Research Computing group using grant numbers NUSREC-HPC-00001”